

Assembly Induced Super-Large Red-Shifted Absorption: The Burgeoning Field of Organic Near-Infrared Materials

Luyang Zhao¹, Xiaokang Ren^{1,2} & Xuehai Yan^{1,2,3*}

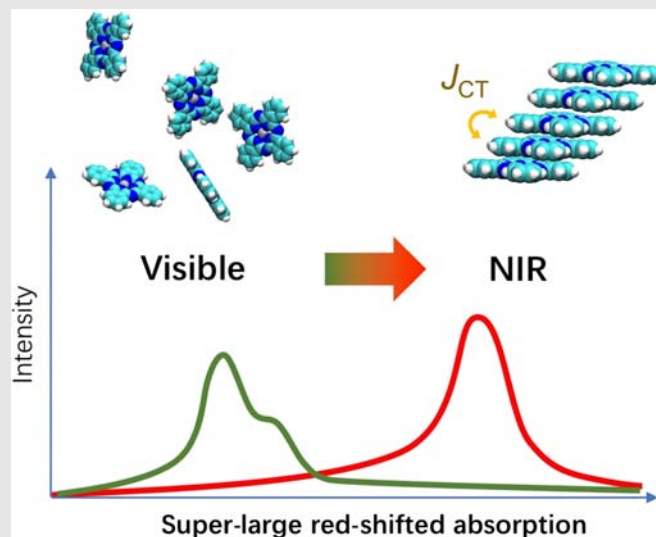
¹State Key Laboratory of Biochemical Engineering, Institute of Process Engineering, Chinese Academy of Sciences, Beijing 100190, ²School of Chemical Engineering, University of Chinese Academy of Sciences, Beijing 100049, ³Center for Mesoscience, Institute of Process Engineering, Chinese Academy of Sciences, Beijing 100190

*Corresponding author: yanxh@ipe.ac.cn

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Supramolecular assembly of organic dye compounds with *J*-aggregation leads to a red-shifted absorption spectrum that greatly facilitates the construction of near-infrared (NIR) materials. A considerable improvement of the material functions requires that the absorption red-shift be larger than 100 nm, but such a super-large red-shift is challenging, and the rules leading to the super-large red-shifted absorption is still not explicit. In this review, we focused on those *J*-aggregated organic dye materials with super-large red-shifted absorption. The nature of the super-large red-shift is originated from the intermolecular charge transfer between neighboring chromophores. The super-large red-shift can be obtained by tuning either the molecular structure or kinetic assembly process in a delicate manner. Materials with super-large red-shifted absorption have been successfully applied to biological imaging, phototherapy, electronic devices, and solar cells, and show great potential in many other fields. The elaboration of assembly induced super-large red-shifted absorption is promising for design of supramolecular NIR materials with

tuned structures, enhanced functionalities, and a wide array of applications.



Keywords: super-large red-shifted absorption, intermolecular charge transfer, supramolecular assembly, kinetic self-assembly, organic near-infrared materials

Introduction

Organic dye materials with near-infrared (NIR) absorption (i.e., NIR materials or more specifically, NIR

absorption materials) are highly attractive in many fields due to their unique advantages of low cost, narrow band gaps, tunable energy levels, and high biocompatibility.^{1–5} Supramolecular assembly of organic dye molecules may

lead to *J*-aggregate states with red-shifted absorption, which has the potential to reach the NIR region in a much more convenient and flexible way than conventional molecular design and synthesis. Ideally, for a dye compound with absorption in the far-red region, a larger than 100 nm of absorption red-shift, which we defined as super-large red-shift,⁶ is required to achieve the NIR region. However, conventional *J*-aggregates merely gave rise to less than 30 nm of absorption red-shift,^{7–9} resulting in quite limited improvement in material properties. How to enlarge the assembly induced absorption red-shift to more than 100 nm becomes the key task for developing supramolecular NIR materials.

To date, there have been few reports of developing super-large red-shifted absorption materials.^{10,11} These materials exhibited larger than 100 nm or even 200 nm of red-shift in comparison with their assembly units, enabling the absorption of the final materials strongly expanding to the NIR region. Consequently, these materials show the feasibility to fabricate NIR materials via simple supramolecular assembly. Despite these successes, assembly methods for these super-large red-shifted absorptions differ from case to case, and assembly units have special molecular structures where the relationship between the molecular structure and the super-large red-shift is still not explicit. In other words, a general rule that governs the fabrication of super-large red-shifted absorption material is still deficient. It is extremely important to reveal the inherent mechanisms of super-large red-shifted absorption for further development of supramolecular NIR materials.

In this review, we focus on the assembly induced super-large red-shifted absorption of organic dye compounds and their various applications. We first elaborate the intermolecular aggregation theory beyond the conventional Kasha model, in which the absorption red/blue shifts are controlled by the intermolecular charge transfer (CT). Next, we summarize the kinetic assembly methods that may lead to the aggregation state with super-large red-shifted absorption. Finally, we highlight the applications of the materials with super-large red-shifted absorption. We believe that this timely review will prompt the development of supramolecular chemistry as well as their prospective applications.

Theory

Conventional *J*-aggregates with small red-shifted absorption have been well explained by Kasha's theory.¹² It assumes that the intermolecular interactions between neighboring chromophores only include the long-range Coulombic coupling, which arises from the interaction between the local excited transition dipole moments (μ) of each chromophore (Figure 1a). The electron configuration under Coulombic coupling is depicted in Figure 1b.

On the one hand, the Coulombic coupling determines the excitation energy by the equation

$$E_F(k) = E_{S_1} + 2J_{\text{Coul}} \cos k \quad (1)$$

where E_{S_1} is the excitation energy of the monomeric chromophores, and k a phase parameter. On the other hand, the Coulombic coupling is dependent on the geometry of the chromophore aggregates with the relationship

$$J_{\text{Coul}} = \frac{\mu^2(1-3\cos^2\theta)}{4\pi R^3} \quad (2)$$

where R is the interchromophore distance, and θ is the angle between μ and R . As a total consequence, the relative energy between the chromophore aggregates and monomers, that is, $[E_F(k) - E_{S_1}]$, crucially relies on the chromophore geometries. When J_{Coul} is negative, the excitation energy of the aggregates is smaller than the monomers, resulting in a red-shifted absorption; the aggregates are slipped "head-to-tail" and are defined as *J*-aggregates. Comparably, when J_{Coul} is positive, the excitation energy of the aggregates is larger than the monomers, resulting in a blue-shifted absorption, and the corresponding aggregate geometries are defined as *H*-aggregates. Since Coulomb coupling is a weak intermolecular interaction "through space", its perturbation over the energy level of S_1 is slight, and the absorption red- or blue-shift is thereby relatively small. The Kasha's theory based on the Frenkel exciton model has shown great success in describing those systems in which chromophores stay far from each other (usually >5 Å) or chromophores are disorderly arranged.

However, Kasha's theory fails when chromophores get close to each other (usually <5 Å, and especially at ~3.5 Å, the distance of π - π stacking), as widely seen in crystallized one-dimensional fibrils, because such a geometry allows wave function overlap between neighboring chromophores. The wave function overlap is a short-range superexchange interaction that contrasts with the Coulombic interaction, enabling the dissociation of local excitations into CT states where electrons and holes reside on neighboring chromophores (Figure 1b).¹³ The effect of the short-range exciton coupling is given by

$$J_{\text{CT}} = -2 \frac{t_e t_h}{E_{\text{CT}} - E_{S_1}} \quad |E_{\text{CT}} - E_{S_1}| \gg |J_{\text{Coul}}|, |t_e|, |t_h| \quad (3)$$

where E_{CT} is the energy of the CT state, and t_e , t_h are the electron/hole transfer integrals, respectively, with the definitions

$$t_e \equiv \langle \phi_L^1 | \mathbf{h} | \phi_L^2 \rangle \quad (4)$$

$$t_h \equiv \langle -\phi_H^1 | \mathbf{h} | \phi_H^2 \rangle \quad (5)$$

in which $\mathbf{h}$ is the single-electron Hamiltonian operator. Similar to the role of J_{Coul} in the Frenkel exciton model,

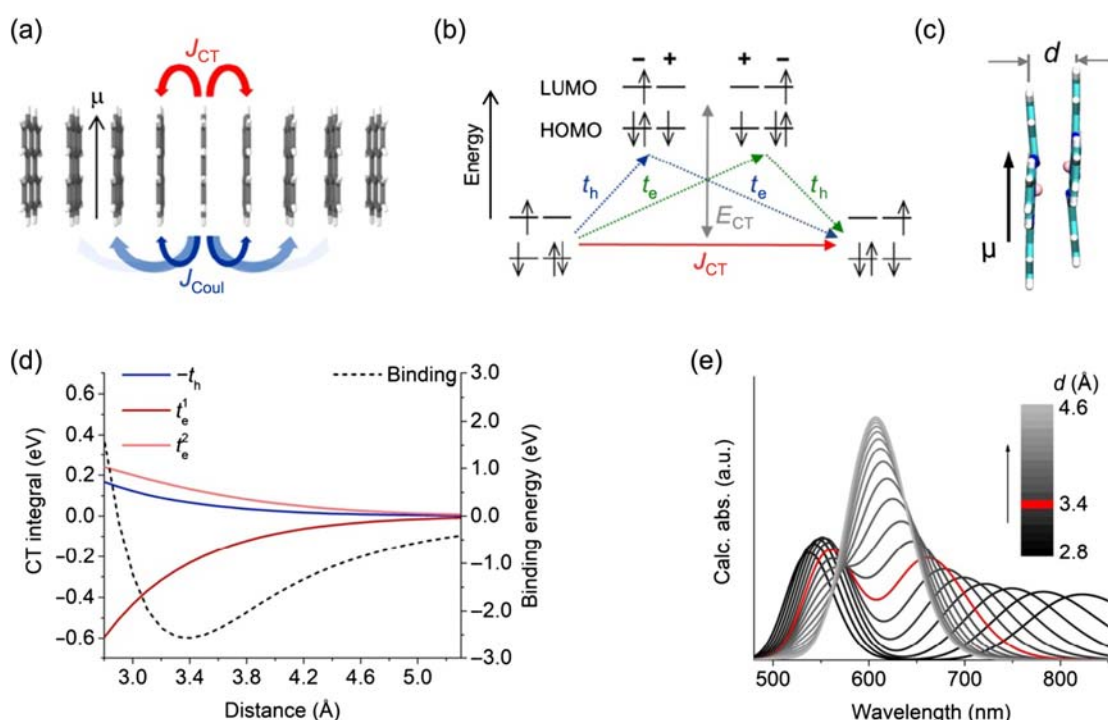


Figure 1 | (a) Schematic illustration of long-range interaction (J_{Coul}) and short-range interaction (J_{CT}). (b) Energy-level diagram depicting the electronic configuration and J_{CT} . Adapted with permission from ref 13. Copyright 2017 American Chemical Society. (c) The stacking model of a phthalocyanine dimer with intermolecular distance d . (d) CT-related integrals as the functions of d . (e) Calculated absorption spectra of the phthalocyanine dimer with different d values. Adapted with permission from ref 6. Copyright 2019 Chinese Chemical Society.

the sign of J_{CT} in the CT model determines whether the absorption spectrum appears red- or blue-shift, and the detailed relationship is

$$E_-(k) = E_{S_1} - 2 \frac{t_e^2 + t_h^2}{E_{CT} - E_{S_1}} + 2J_{CT} \cos k \quad (6)$$

$$E_+(k) = E_{CT} + 2 \frac{t_e^2 + t_h^2}{E_{CT} - E_{S_1}} - 2J_{CT} \cos k \quad (7)$$

under the perturbation limit of $J_{Coul} = 0$.¹⁴ Although the interference between Coulombic and CT-mediated coupling makes the mathematics more complicated, attention should be paid to the qualitative inferences drawn from the CT-mediated exciton model. Due to the existence of t_e and t_h , the energy level of the excited state is drastically perturbed, leading to severe spectral variation from which the super-large red-shifted absorption stems. Clearly, the difference between super-large red-shift and conventional small red-shift is that the former has remarkable electron/hole transfer in addition to the larger shift scale. Moreover, it is easy to investigate the relationship between supramolecular structure and super-large red-shifted absorption from two independent dimensions, that is, the vertical and planar dimensions, as follows.

The effect of vertical intermolecular distance

As mentioned earlier, CT coupling is a short-range interaction, and the intermolecular distance has significant influences on the CT coupling strength. This is because both t_e and t_h decay exponentially with the intermolecular distance. Taking the distance along the direction vertical to the π -plane of the chromophore, the influence of intermolecular distance on the absorption red-shift will be clearer. Taking the phthalocyanine dimer as an example, we calculated the relationship between its vertical distance and the absorption spectrum (Figure 1c).⁶ The geometry of the phthalocyanine dimer was deduced from the experimental results and optimized to an energy minimum. The t_e and t_h were evaluated by the frontier orbital energy splitting with the relationships $t_e = 2\Delta E_L$ and $t_h = -2\Delta E_H$, where ΔE_L was the energy difference of the lowest unoccupied molecular orbitals (LUMOs) and ΔE_H was the energy difference of the highest occupied molecular orbitals (HOMOs). There were two t_e s, namely t_e^1 and t_e^2 , because monomeric phthalocyanine has two degenerated LUMOs. Accordingly, both t_e and t_h decreased dramatically with increasing intermolecular distance (Figure 1d). When the distance was larger than 5.0 Å, the CT coupling almost diminished, leaving the Coulombic coupling effect alone, and the corresponding

spectrum merely showed a small blue-shift (Figure 1e). In contrast, when the distance was close to 3.4 Å, the geometric energy-minimal distance, t_e and t_h were large, resulting in a significant absorption red-shift. The remarkable split of absorption in the Q-band region was caused by the two t_e s with opposite signs, which lead to a relatively large red-shift and a relatively small blue-shift.

The effect of in-plane slip

In addition to the vertical distance, intermolecular displacement (or the slip to the π -plane of the chromophore) also significantly affects the supramolecular absorption spectrum because the π -conjugated chromophores have nodal frontier orbitals, and the slide geometry directly influences the orbital overlapping.¹⁵ Taking perylene as the example, the nodal patterns of HOMO and LUMO of the perylene molecule determine the sign and the magnitude of t_h and t_e , which further leads to the transverse displacement-dependent $t_e t_h$ ($\propto J_{CT}$).¹⁶ The $t_e t_h$ is extremely sensitive to the intermolecular slide so that an absorption red-shift can be obtained only if the supramolecular geometry of the aggregates is located in the adequate region with large and negative $t_e t_h$. Comparably, the Coulombic coupling is less sensitive to the intermolecular displacement, because along with the long molecular axis, the J_{Coul} suffers only one conversion from positive to negative, which corresponds to the *H*- and *J*-aggregate patterns, respectively.

The effect of in-plane slide on the absorption was also manifested by a cyanine derivative (3,3'-bis(4-sulfobutyl)-5,5'-dichloro-11-diphenylamino-10,12-ethylenethiatricarbocyanine, ammonium salt, Cy7-DPA) dye, as investigated by Caram and co-workers.¹⁷ The Cy7-DPA monomers had intrinsic absorption at 807 nm. In comparison, their self-assembled two-dimensional aggregates showed super-large red-shifted absorption lying at ~1050 nm. The slip between neighboring chromophores of this aggregates was estimated to be 7~10 Å, which was correlated to 0.36 times of the chromophore length (Figure 2a). Such a long slip was considered reasonable because the steric diphenylamino group on Cy7-DPA was too large to allow a slip less than 7 Å. Furthermore, the calculation of density of states (DOS) and bright state as the functions of intermolecular slip indicated that along with the slip increasing, the aggregates would alternately exhibit *H*- and *J*-type geometries with corresponding blue- and red-shifted absorption (Figure 2b). The experimentally obtained aggregates with super-large red-shifted absorption was just located on one of the slip regions of *J*-type geometries. Moreover, an *I*-type geometry that fell between *H*- and *J*-type aggregates was identified, which showed unique temperature-dependent absorption spectrum and was probably the "Null"-aggregate referred by the CT exciton theory with $J_{Coul} + J_{CT} = 0$.¹⁸ Although this investigation was depicted using the language of

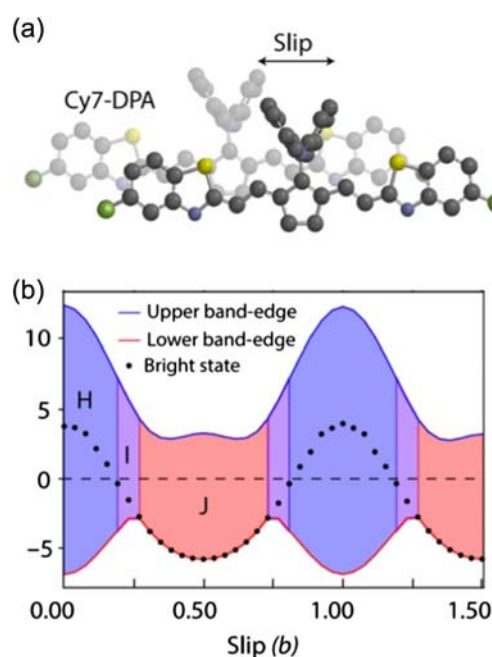


Figure 2 | (a) Predicted molecular stacking pattern of the Cy7-DPA dye. (b) Bright states and the blue-/red-shift alternation as the function of slip, where *b* refers to the molecular length. Adapted with permission from ref 17. Copyright 2019 American Chemical Society.

Kasha's theory, it definitely represented the picture of CT coupling.

As a short summary of this section, a few principles that lead to the super-large absorption red-shift could be drawn from the complicated mathematics of CT exciton model. Namely, the assembled chromophores should (1) have very short intermolecular distances (in general, ~3.5 Å) with neighboring molecules, and (2) achieve an intermolecular transverse displacement with a strong negative CT coupling effect. Due to these requirements, an ordered stacking pattern of the building blocks is usually necessary to achieve the super-large red-shifted absorption, and a more precise control over both the molecular structure of the building block and the assembly process is also indispensable.

Strategies for Obtaining Super-Large Red-Shift

Based on the CT exciton model and quantum chemistry calculation, one can predict whether there is a supramolecular structure exhibiting the super-large red-shifted absorption starting from a certain molecular building block. However, these structures with super-large red-shifted absorption are not always thermodynamically or kinetically accessible. To obtain the super-large red-shifted absorption, extraordinary efforts need to be

made on the molecular structure of the building blocks and the supramolecular assembly process.

Molecular modification of substitute groups

Organic dye compounds are generally constituted by a π -conjugated chromophore and peripheral substitutes. While it is difficult to form the aggregation pattern with super-large red-shifted absorption from a single chromophore, it is possible to tune the thermodynamics of the aggregates via substituent couplings so that the aggregate geometries can locate in the adequate region.¹⁹ The effect of substituents on the absorption spectra variance was represented by Würthner and co-workers.²⁰ They used a series of dye compounds with an identical backbone of conjugated 2-aminothiophene and 2-[4-(*tert*-butyl)thiazol-2(3*H*)-ylidene]malononitrile and variable substituents on the nitrogen site, including several kinds of alkyl and phenyl groups. The monomeric solution of all these compounds exhibited similar absorption with maxima at 649~664 nm. In contrast, the solid states of these compounds prepared by spin coating from chloroform solution exhibited either blue- or/and red-shifted absorptions with maxima at 479~489 nm (*H*-band) and 735~751 nm (*J*-band), respectively (Figure 3). For substituents with low steric demand and/or rigidity, a tight card stack-like packing with no longitudinal displacement can be adopted in the solid state, yielding the largest dipole-dipole interaction and the strongest *H*-coupling of the dyes' transition dipole moments. In

contrast, for larger, flexible, or bulky aromatic substituents, where such a cofacial arrangement is disfavored due to the increased steric demand, a zig-zag packing is induced in the solid state by large longitudinal shifts within the dimer synthon, thus giving rise to *J*-coupling.

The effect of substituents was active in various chromophores. Jiang and co-workers²¹ found that long-chain substituents in phthalocyanine could induce a *J*-aggregation with super-large red-shifted absorption. Generally, substituent-free phthalocyanines tended to form *H*-aggregations due to strong face-to-face π - π interactions. With the addition of hydrophilic-substituted groups which were covalently bonded to the central coordinated silicon ion, the axially substituted silicon phthalocyanine preferentially formed *J*-aggregations with ca. 100 nm of absorption red-shift. The axial-substituted groups were revealed to possess a *cis*-conformation, which sterically hindered the face-to-face aggregation but facilitated a head-to-head aggregation. Interestingly, the substituents should form a long chain; otherwise, the *cis*-conformation cannot remain stable.

The aggregation structure and the absorption spectrum may also be synergistically controlled by multiple substituents on one chromophore backbone. For example, naturally derived light-harvesting chlorins exhibit fantastic assembly features with possibly as large as ~90 nm of bathochromic shifted absorption from the far-red region to the NIR region. However, the large red-shift relies on three-substituted groups on the chlorin structure: (1) the central Zn ion coordination; (2) 3'-methoxy (or hydroxyl) substitution; and (3) a steric hydrophobic ester on the 17-carboxylic group. The chlorin derivative could not form the aggregate with the large red-shifted absorption unless all of the three groups were contained.²² This is because the tetra-coordinated Zn ion enabled extra axial coordination with the electro-negative oxygen on the 3'-site of the neighboring molecule, leading to a long-range ordered and slipped π -stacking arrangement.²³ In addition, the steric hydrophobic group probably further stabilized the large intermolecularly slipped geometry. Therefore, rational design of molecular structure is important for achieving the supramolecular aggregates with the desired red-shifted absorption spectra.

Supramolecular assembly via intermolecular interactions

Sometimes two or more thermodynamically accessible aggregates with local minimal energy in the energy landscape can be obtained from the same building blocks. The super-large red-shifted absorption may be hidden in one of these aggregates but is difficult to achieve via conventional assembly methods like rapid precipitation due to the pathway complexity. Achieving such aggregation states therefore requires kinetic control over the

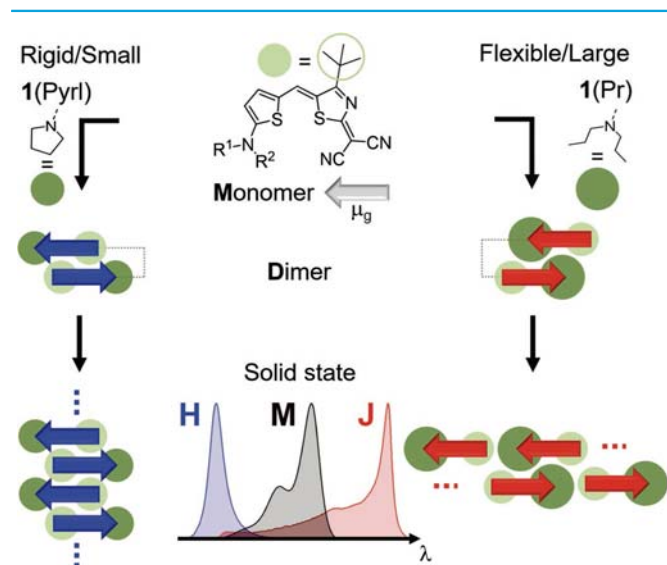


Figure 3 | Rigid and small peripheral substitutes lead to *H*-aggregates of the merocyanine chromophore with blue-shifted absorption, while flexible and large peripheral substitutes lead to *J*-aggregates with red-shifted absorption. Adapted with permission from ref 20. Copyright 2018 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.

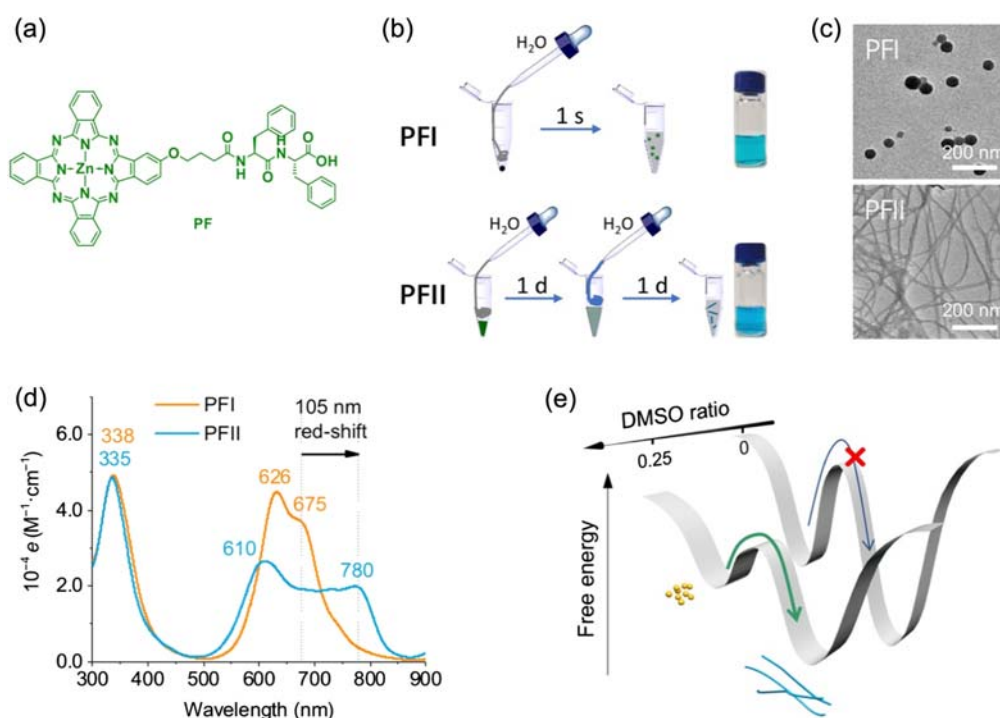


Figure 4 | Kinetically controlled supramolecular assembly enabling super-large red-shifted absorption. (a) The molecular structure of the assembly unit PF. (b) Two assembly processes. (c) Assembly morphologies of PFI and PFII resulting from the different assembly process. (d) The absorption spectra of PFI and PFII. (e) The schematic free-energy landscape of PF assemblies. Adapted with permission from ref 6. Copyright 2019 Chinese Chemical Society.

assembly process, or more precisely, control of noncovalent intermolecular interactions in the dimensions of both time and space via several parameters, including temperature, external electric or magnetic field, solvent pH, the order of addition, and so forth.^{24–27} With suitable kinetic methodologies, different aggregate morphologies and molecular arrangements can be rationally selected from the same starting building blocks.^{28–30}

Kinetic control of self-assembly

Recently, we showed a pair of pathway-dependent assembly structures with and without super-large red-shifted absorption, where a short peptide decorated phthalocyanine (phthalocyanine-diphenylalanine, PF) was employed as the model assembly unit (Figure 4a).⁶ Through conventional flash precipitation, which was an entropy-dominated assembly process, the phthalocyanines assembled to nanoparticles (PF nanoparticles, PFI) with the absorption of maximal-wavelength at 675 nm, which was the same as the monomeric molecules and at ~630 nm as a blue-shifted absorption probably due to *H*-aggregation. In contrast, slowly adding water many times over 3 days to elongate the PF assembly process afforded fibril nanoassemblies (PF nanofibrils, PFII) with a 105 nm red-shifted absorption, remarkably expanded to the NIR region (Figures 4b–4d). Further experiments

revealed that the PFII aggregates were thermodynamically more stable than PFI, but there was a large enough energy gap between these two assembly states such that conventional flash precipitation could not lead to PFII. However, increasing the starting volume ratio of dimethyl sulfoxide (as the good solvent to solve the phthalocyanine) would reduce the energy gap, thereby increasing the accessibility of PFII (Figure 4e).

A similar supramolecular assembly system that was controlled by solubility and assembly time to exhibit super-large red-shifted absorption was also found in a BF₂-chelated azadiapyrromethene (aza-BODIPY) system.³¹ Kinetic assembly of the aza-BODIPY monomers led to metastable nanoparticles with small red-shifted absorption, which then slowly transformed to thermodynamically more stable rod-like micelles with sharp and super-large red-shifted absorption. The nanoparticle and the micelle as two different aggregates constitute a moderate energy barrier, thereby enabling the spontaneous transformation. Herein, both the aza-BODIPY and the aforementioned phthalocyanine assemblies with super-large red-shifted absorption was thermodynamically more stable in the energy landscape, while their preparations require a special kinetic assembly process.

Notably, the controls over the kinetic assembly processes and over the molecular structure may be synergistic for the super-large red-shifted absorption. Zhao

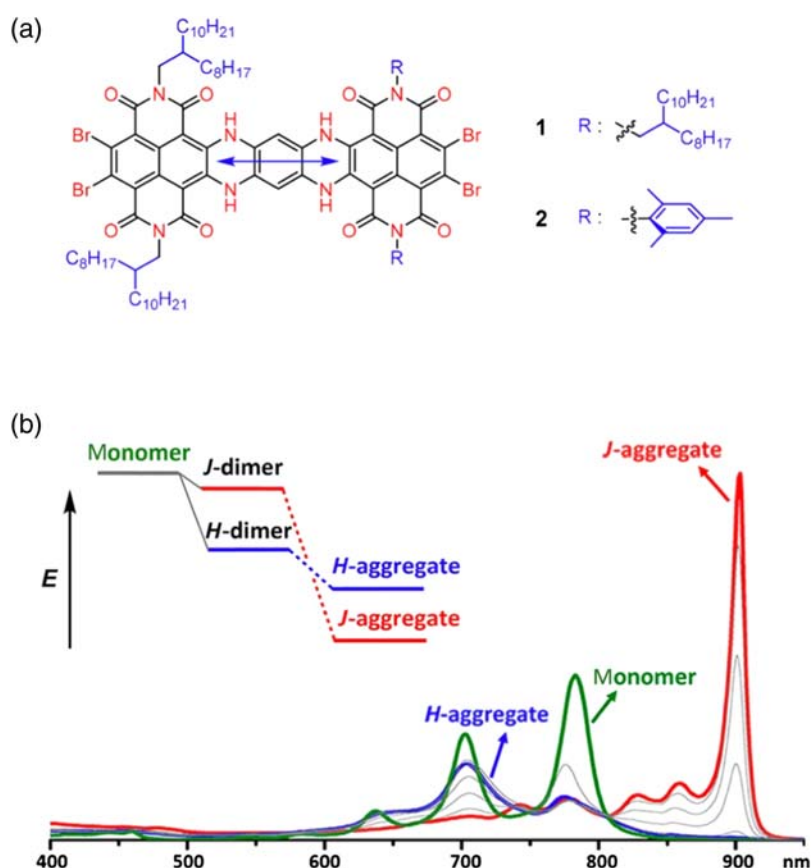


Figure 5 | (a) Molecular structures. (b) Absorption spectra of the monomer, H-aggregate, and J-aggregate of compound 1. The inset is the schematic energy level diagram of different assembly states of 1. Adapted with permission from ref 32 Copyright 2018 American Chemical Society.

and co-workers³² showed that an *N*-heterocyclic aromatic dicarboximide molecule with four long alkyl substitutes exhibited maximal absorption wavelengths at 770 nm as monomers and at 900 nm as thermodynamically stable *J*-aggregates (Figures 5a and 5b). The *J*-aggregates could feasibly be obtained by either lowering the solution temperature or increasing the nonpolar solvent ratio. However, other relatively stable *H*-aggregates that appeared as competing species were able to intercept the *J*-aggregation path, representing off-pathway assembly states. In the *H*-aggregates, there is ample space on both sides of the stacked molecules to accommodate these alkyl chains. As the size of *H*-aggregates grew larger and larger, the void space available for each molecule diminishes, and steric congestion continuously built up. Reasonably, when two of the β -branched alkyl chains were changed by mesityl groups with low steric flexibility as well as solubility, *H*-aggregate formation was frustrated, and only the *J*-aggregates were observable, shown by the prominent *J*-band at 874 nm. Since both of these two molecules showed similar super-large red-shifted absorption as *J*-aggregates, the change of substitutes only varied the assembly process but not the aggregation pattern.

Template-assisted self-assembly

In addition to the kinetically controlled self-assembly of individual molecular compounds, external species-participated assembly could alternatively lead to the special assembly states with super-large red-shifted absorption. A cyanine dye, IR140-Cy⁺, is soluble in alcohols with absorption maximum at 800 nm and insoluble in water.³³ Directly adding the IR140-Cy⁺ to water or to an aqueous glutathione (GSH) solution gave rise to *H*-aggregates with a broad blue-shifted absorption spectrum. Comparably, adding the IR140-Cy⁺ to the solution of GSH-capped PbS quantum dots, a *J*-aggregation of the dye was formed, showing a super-large red-shifted absorption to 965 or 1040 nm. The *J*-aggregation of IR140-Cy⁺ on the quantum dots was thought to be controlled by the electrostatic interaction between the positively charged dyes and the negatively charged GSH groups on the quantum dots surface. Such *J*-aggregates with two red-shifted absorption maxima were further found affected by the pH, which was probably because the hydrogen-bonding interaction other than the electrostatic interaction played a role in the supramolecular structure.

In many cases, a photosensitive compound could simply form *J*-aggregation with the help of an amphiphilic polymer, which was also known as polymer encapsulation.³⁴ Indocyanine green (ICG) showed intrinsic absorption peaks at 713 and 780 nm. After coassembly with polymers like 1,2-distearoyl-*sn*-glycero-3-phosphoethanolamine-*N*-[amino(polyethylene glycol)-2000] (DSPE-PEG₂₀₀₀-NH₂) or poly(lactide-co-glycolide-*b*-poly(ethylene glycol)) (PLGA-*b*-PEG), the ICG nanoparticles showed a significantly red-shifted peak at 895 nm.^{35,36} In a similar manner, coassembly of an aza-BODIPY compound and amphiphilic DSPE-PEG₂₀₀₀ under ultrasonic power afforded cubic nanoparticles showing two newly appearing absorption peaks at 721 and 772 nm, while the aza-BODIPY monomer and its *H*-aggregates showed absorptions at 658 and 624 nm, respectively.³⁷ Coassembly of an amphiphilic polymer H-PEG-PPG-PEG-OH and zinc phthalocyanine lead to a *J*-aggregate with the absorption peak red-shifting from 680 to ca. 800 nm.³⁸ With the help of the emulsification process, coassembly of poly(lactic-co-glycolic acid) (PLGA), *n*-hexadecylamine-substituted hyaluronic acid, and zinc phthalocyanine lead to a larger absorption red-shift from ca. 680 to 832 nm.³⁹ Among all of these cases, the amphiphilic polymers were essential for the super-large red-shifted absorption. Despite the lack of detailed assembly mechanism analysis, it is reasonable to conclude that intermolecular interactions played a critical role in the formation of *J*-aggregates.

The external species that assisted the assembly could be covalently conjugated to the assembling chromophores. For example, encapsulation of a chlorin derivative, zinc methoxypropheophorbide *a* (Zn-MeO-chlorin acid), to dipalmitoylphosphatidylcholine (DPPC) and a PEGylated phospholipid afforded biomimetic lipid nanovesicle membranes with their absorption spectrum quite similar to molecular Zn-MeO-chlorin acid. While 1-palmitoyl-2-hydroxy-*sn*-glycero-3-phosphocholine (PHPC) was conjugated by the Zn-MeO-chlorin acid to form Zn-MeO-chlorin lipid, the resultant nanovesicles exhibited 72 nm of red-shifted absorption.²² The remarkable red-shift was considered from the ordered assembly structure of Zn-MeO-chlorin, and the structure was probably originally induced by the lipid-lipid interactions between PHPC and DPPC. Intense circular dichroism signals of the Zn-MeO-chlorin lipid further confirmed the existence of the ordered assembly structures.

Applications

Biological imaging and phototherapy

There are several unique advantages of NIR materials for biological purposes: (1) they enable a higher imaging sensitivity because living cells or tissues have low absorption and emission in the NIR region; (2) they reduce

the tissue damage from irradiation; and (3) they allow deeper tissue penetration than visible light.^{40–42} These advantages will be more prominent for NIR-II materials, or short-wave infrared materials.⁴³ To achieve such a low-energy absorption for efficient biological applications, supramolecular assembly of organic chromophores to acquire a super-large red-shifted absorption in the NIR-II region has proved to be a more promising route than molecular design.

Fluorescence and photoacoustic imaging

The photoacoustic effect is based on the optical absorption that generates acoustic waves, and fluorescence is based on the emission. Challenges that remain for the application to biological imaging are the material stability and delivery efficiency. The commonly used supramolecular assembly method that leads to both a super-large red-shifted absorption and high delivery efficiency is the polymer encapsulation.³⁴ This method has been introduced in a few examples mentioned earlier. However, to properly convert these assemblies into bioavailable materials, more work needs to be done. Li and co-workers⁴⁴ reported a croconaine dye A1094 with intrinsic absorption at 1094 nm and a super-large red-shifted absorption at ca. 1230 nm under the encapsulation of DSPE-PEG₂₀₀₀. Although the *J*-aggregate could be formed under a wide range of molar ratios between the dye and the polymer, a 1:1 ratio was finally chosen as the adequate particle size and dye-loading efficiency. The polymeric nanoparticles were intravenously injected and showed a maximal tumor accumulation at 36 h (detected by fluorescence), and a maximal photoacoustic signal (970 and 1200 nm) at 8 h, demonstrating both the biosafety and efficient photoacoustic imaging ability.

In addition to the polymeric encapsulation, other methods are also available for the bioimaging. Sletten and co-workers⁴⁵ successfully employed hollow mesoporous silica nanoparticles (HMSNs) to load the IR-140 photosensitizer (Figure 6a). As mentioned earlier, the dye IR-140 formed a special *J*-aggregation under the guidance of a quantum dot template enabling a larger than 165 nm of absorption red-shift to the NIR-II region. However, the single-quantum dots obviously did not accommodate to the biological living systems. By treating HMSN with IR-140, *J*-aggregation with J1 and J2 absorption bands at 965 and 1040 nm was formed, and the aggregates were confirmed mainly being protected inside the HMSN cavity (Figures 6b and 6c). Importantly, the HMSN-loaded *J*-aggregates kept ~90% maintenance in phosphate-buffered saline (PBS) solution for 2 weeks, while only ~8% *J*-aggregates in solution were maintained for 1 day, suggesting the significant role of HMSN in stabilizing the *J*-aggregation. In a similar way, the HMSN-loaded *J*-aggregates also showed higher photostability than the *J*-aggregates in solution and the

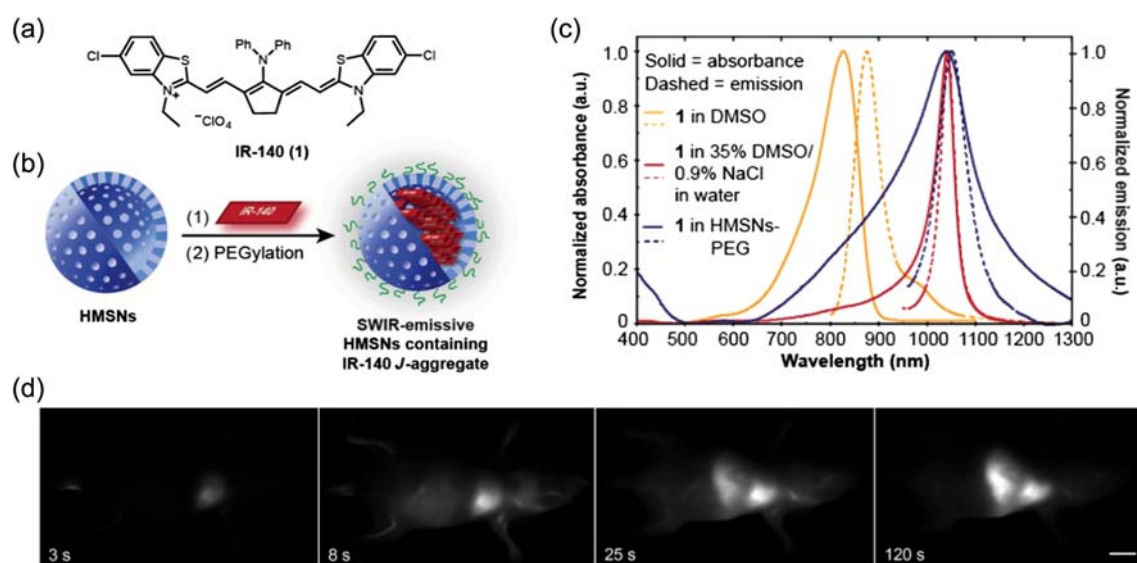


Figure 6 | (a) Molecular structure of IR-140 (**1**). (b) Schematic illustration of the assembly strategy. (c) Normalized absorption and emission spectra of IR-140 monomer, J-aggregate in solution, and J-aggregate in HMSN. (d) In vivo imaging at 16 fps (980 nm, 91 mW cm² excitation; 1000–1700 nm collection) upon i.v. delivery of HMSN-loaded IR-140 J-aggregate. Adapted with permission from ref 45. Copyright 2019 American Chemical Society.

monomers. Due to the excellent stability and biocompatibility provided by the HMSN cargo, the dye-loading HMSNs enabled intravenous injection into nude mice and exhibited intense signal at different organs under the excitation at 980 nm and collection on 1000–1700 nm (Figure 6d).

It is highly desirable to employ nature-derived compounds for biological purposes due to their excellent biocompatibility. Therefore, developing supramolecular nature-derived materials with super-large red-shifted absorption appears significant. As mentioned earlier, Chlorophyll derivatives with central zinc complexation and 3¹-methoxy substitution may afford special J-aggregates with remarkable red-shifted absorption, but the assembly usually requires external templating species. Encapsulation of phorpholipid-conjugated bacteriopheophorbide a to the liposome successfully provided the desired NIR absorption at 824 nm in maximal with 72 nm of red-shift. Such J-aggregates were also fabricated by using lipoproteins, microbubbles, and chlorosome mimetics, all of which enabled photoacoustic imaging in the corresponding imaging wavelength with considerable in vivo stability.⁴⁶

Phototherapy

Phototherapy is divided into photodynamic therapy and photothermal therapy. Photodynamic therapy requires a high singlet oxygen quantum yield (Φ_{Δ}) while photothermal therapy requires a high light-to-heat conversion efficiency. Amphiphilic BODIPY derivative compounds encapsulated by polymers with 20% loading efficiency exhibited enhanced and red-shifted absorption band at 770 nm compared with the absorption of the BODIPY

monomer at 665 nm.^{47,48} The super-large red-shift was possibly owing to the formation of J-aggregates with distinct π - π cofacial stacking. The vesicles that contained the J-aggregates exhibited poor singlet oxygen yield but profound photothermal conversion efficiency under the 785 nm laser irradiation, which was in contrast with their performance under 660 nm irradiation. After intravenous injection of the BODIPY J-aggregate vesicles and followed by 785 nm irradiation on the tumors for 24 h, a potent hyperthermia ($\Delta T \approx 15.0$ °C) was triggered, which further lead to effective tumor ablation through destructive necrosis. The red-shifted absorption rendered 785 nm laser irradiation for treating the tumor.

The activation of singlet oxygen requires that the light wavelength be shorter than 1270 nm for photodynamic therapy. Comparably, there is no such limitation for photothermal therapy. Therefore, NIR-II materials have far superior advantages in photothermal therapy. Squaraine is a widely used fluorescent dye easy that easily forms J-aggregates in aqueous solution. Fan and co-workers⁴⁹ synthesized a bispyrrole-conjugated squaraine (SQP) molecular compound with intrinsic absorption and emission both at the NIR-I region (approximately 800 nm). The SQP assembly was pathway-dependent, and the J-aggregation was achieved by polymer encapsulation, which showed an intense absorption at 901 nm and a broad emission band extending to 1400 nm.

Optoelectronic materials

Charge-carrier mobility and exciton diffusion are the key parameters that determine the performance of organic

semiconductors in optoelectronic applications.⁵⁰ Supramolecular assembly of small organic molecules directed by noncovalent interactions have been proved facile to tune the charge conductance for recent decades. This is because those prominent close-packing π -extending systems that resulted from the supramolecular assembly enable efficient CT, which is evaluated by t_e and t_h . Since t_e and t_h also determine the mobility and the absorption spectra, it is desirable to screen efficient semiconductive materials from the absorption.

An aggregate with super-large red-shifted absorption may lead to notable electron mobility. Xie and co-workers⁵¹ showed that a perylene bisimide derivative compound could be deposited to crystalline with an intermolecular slip angle of close to 54.7°, the boundary between conventional *H*- and *J*-aggregation. However, the crystalline film still exhibited remarkable absorption red-shift in comparison with its molecular solution, indicating strong CT coupling and hole/electron transfer integral. The crystalline film exhibited typical *n*-type transistor behavior with the electron mobility (μ_e) up to 0.09 cm² V⁻¹ s⁻¹, suggesting an inner correlation between super-large absorption red-shift and the charge-carrier mobility.

The series of peripherally substituted merocyanines mentioned in Figure 3 further showed that the dyes with high hole transport performance are those showing the highest absorption intensity in thin film, including both the dyes **1**(Hex) and **1**(Ph) with large red-shifted absorption, and the dye **1**(Pyr) with large blue-shifted absorption.²⁰ Theoretically, Spano and co-workers¹⁵ showed that absorption spectral line shape and exciton mobility are tightly correlation. Taking the vibration coupling into consideration, if the O-0/O-1 ratio of vibronic oscillator strengths in the solid phase deviates significantly from the ratio observed in the molecular solution phase, a high mobility exciton is yielded. Since the O-0/O-1 ratio of vibronic oscillator strengths is also an indicator of the aggregation pattern, there is potential to use the absorption spectrum to screen for high mobility aggregates.

The hole-transporting materials are also the important component of photovoltaic cells. Taking the perovskite solar cell as an example, which has been proven as an efficient solar cell, the hole-transporting materials extract photogenerated holes from the perovskite material layer and then transport the holes to the counterelectrode. In addition to providing an optimal hole mobility, these materials also possess a little higher energy level of the HOMO than the valence band edge of the perovskite.⁵² Wang and co-workers showed that the film of metal-free chlorophyll derivative (H₂-Chl) with small red-shifted absorption compared with the H₂-Chl solution exhibited low hole mobility. In contrast, the film of zinc chlorophyll derivative (Zn-Chl) with large red-shifted absorption exhibited nearly 10 times higher hole mobility than H₂-Chl.⁵³ The combination of H₂- and Zn-Chl to afford

a bilayer further improved the charge-transporting performance. Here, the super-large red-shifted absorption seemed to have two effects: one is the correlation with high hole transition integral, and the other is to tune the frontier orbital energy levels to match the charge-transporting cascade. The *J*-aggregation-enhanced charge-carrier mobility was also found in a BODIPY derivative, where the hole mobility of the *J*-aggregate of BODIPY with remarkable red-shifted absorption showed more than one order of magnitude higher than that of the *H*-aggregates.⁵⁴ Moreover, Bo and co-workers⁵⁵ fabricated two oligothiophene derivatives, namely, o-4TBC-2F and m-4TBC-2F, and the former formed *J*-aggregation after thermal annealing and showed superior absorption red-shift, resulting in a higher charge-carrier mobility and photoconversion efficiency than the latter material without *J*-aggregation. Although the exquisite relationship between the charge-carrier mobility and super-large red-shifted absorption of a certain aggregate still lacks research, the super-large red-shifted absorption is valuable as the screen of high-performance charge-transporting materials.

Other applications

Super-large red-shifted absorption is a fascinating property and exhibits great potential for a wide range of applications. A melamine and cyanurate complex system was designed to exhibit mutual transformation between *H*- and *J*-type packing with different colors via temperature controlling, realizing external stimuli-responsive optical variation.⁵⁶ A BODIPY nanoassembly was reported to show stimuli-responsive photophysical properties, which would be facily applied in biological probes.^{57,58} These works inspired researchers to discover novel functionalities of these materials.

Conclusions and Perspective

We reviewed the assembly induced super-large red-shifted absorption from physiochemical mechanism to various applications. The super-large red-shifted absorption originated from the closely packed chromophores with specific horizontal slip, which enabled efficient intermolecular CT. Correspondingly, the spectral variation could be explained by CT model, but not by the conventional Frenkel model or Kasha's theory because the latter only considered dipole-dipole intermolecular interaction. Although super-large red-shifted absorption was rarely observed on chromophore aggregates, it could be tuned by either modifying the peripheral substituents on the chromophore or kinetic supramolecular assembly methods. These molecules mentioned in this review are summarized in Table 1. The assembly induced super-large red-shifted absorption showed great potential for application in biological imaging, phototherapy, electronic

Table 1 | Summary of the Molecular Building Blocks that Could Afford Super-Large Red-Shifted Absorption Mentioned in This Review

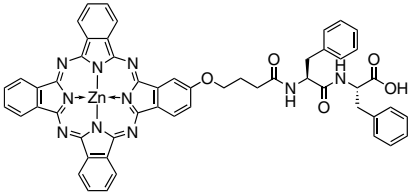
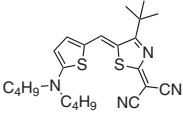
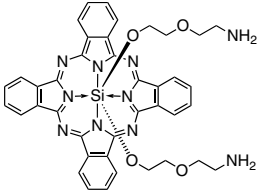
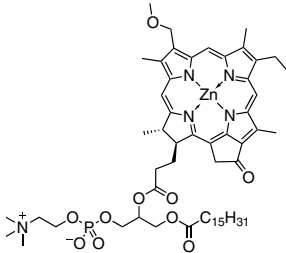
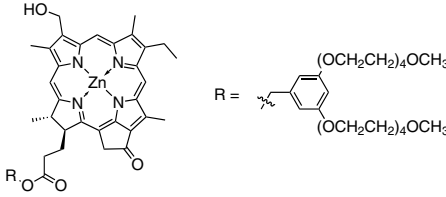
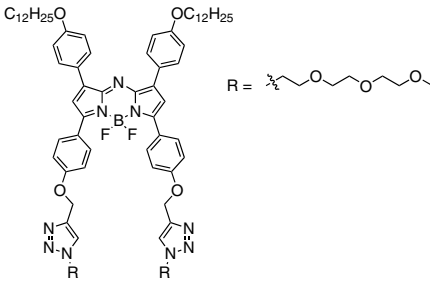
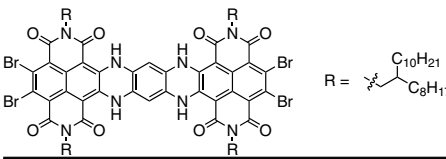
Structure	Molecular Absorption (λ , nm)	Assembly Absorption (λ , nm)	Assembly Method	References
	675	780	Kinetic assembly	6
	654	751	Assembly with tuning molecular structure	20
	674	790	Assembly with tuning molecular structure	21
	653	725	Assembly with tuning molecular structure	22
	656	749	Assembly with tuning molecular structure	23
	688	799	Kinetic assembly	31
	770	900	Kinetic assembly and tuning molecular structure	32

Table 1 (Continued)

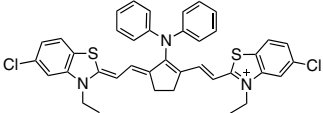
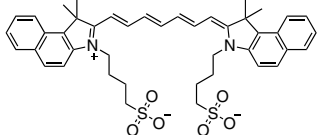
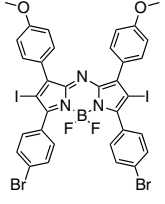
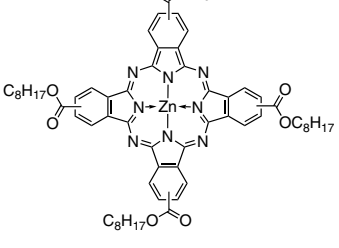
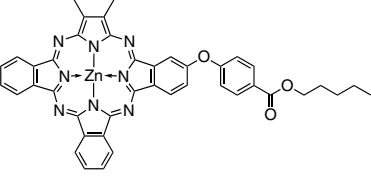
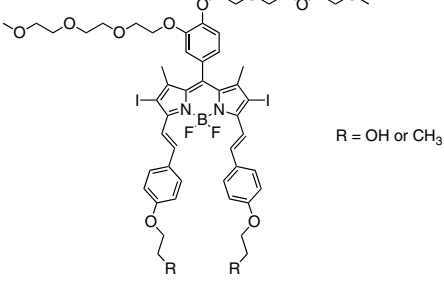
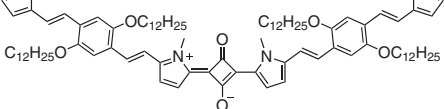
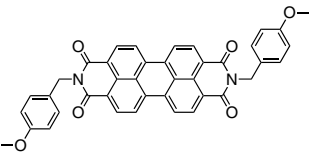
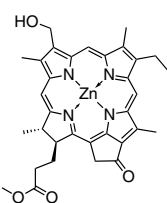
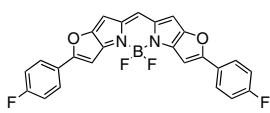
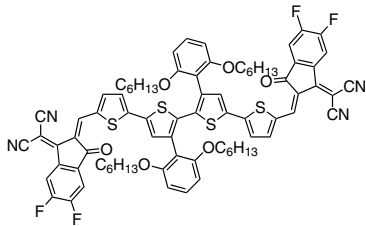
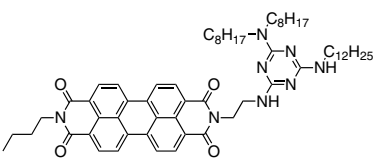
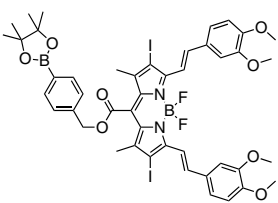
Structure	Molecular Absorption (λ , nm)	Assembly Absorption (λ , nm)	Assembly Method	References
	800	965, 1040	Template-assisted assembly	33, 45
	780	895	Polymer encapsulation	35, 36
	624	721, 772	Polymer encapsulation	37
	~670	ca. 800	Polymer encapsulation	38
	674	832	Polymer encapsulation and emulsification	39
 R = OH or CH ₃	665	770	Polymer encapsulation	47, 48
	766	901	Polymer encapsulation	49

Table 1 (Continued)

Structure	Molecular Absorption (λ , nm)	Assembly Absorption (λ , nm)	Assembly Method	References
	532	680	Physical vapor transfer	51
	647	735	Spin-coating	53
	650	777	Vacuum-deposition	54
	671	825	Spin-coating	55
	515	620	Coassembly with cyanurate guest	56
	708	808	Polymer-templated assembly	57

devices, organic solar cells, and probably many other fields, because it is not only an easy way to obtain NIR absorption, but also enables enhanced charge-carrier mobility. The elaboration of super-large red-shifted absorption will open up a new route for rational design of supramolecular NIR materials with fascinating properties.

Despite these remarkable progresses, a few aspects of this burgeoning field are still in their infancy. First, nearly all the reported systems lack precise supramolecular structure. As a consequence, the explicit relationship between super-large red-shifted absorption and the

structure was unclear. Besides, whether a long-range order of the supramolecular structure is necessary to achieve the super-large red-shifted absorption also remains implicit. Second, a universal method that leads to the super-large red-shifted absorption has not been uncovered. Third, precise theoretical prediction of the super-large red-shifted absorption is urgently needed. Undoubtedly, the solution of these problems will significantly promote the discovery of desired supramolecular NIR materials. In addition, efforts are also expected to pay on increasing the absorption red-shift to larger than

200 nm or more, so that the absorption in the NIR II region is possible, and that the application of the materials would be expanded to many other fields.

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